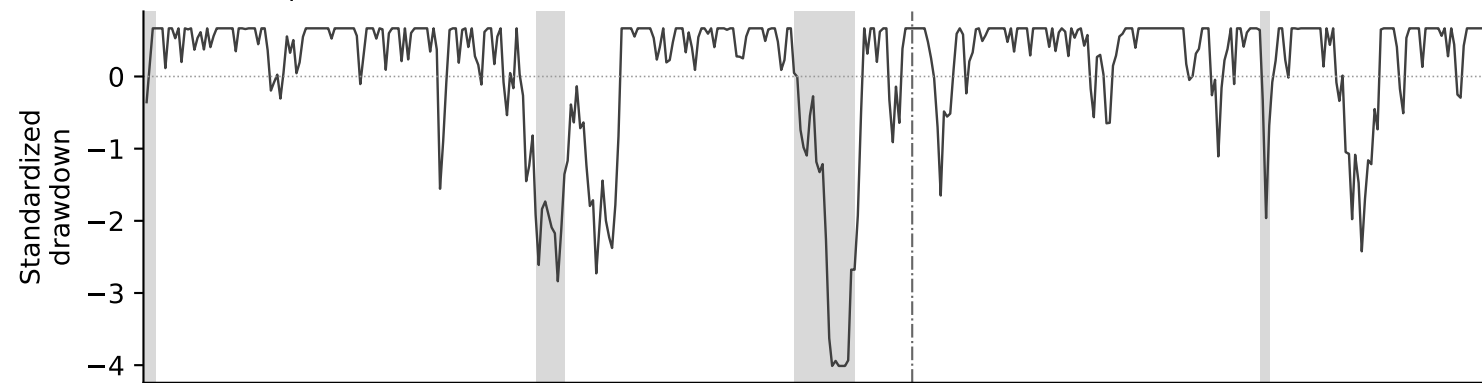


(a) HMM input feature: market drawdown, 1990--2025



(b) Out-of-sample filtered panic probability π_t^{filter} (drawdown-only), 2011--2025

